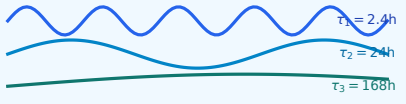
# C-STGB v2: End-to-End Architectural Dataflow & Neuro-Symbolic Tensor Pipeline

*Six Interconnected Modules: Continuous Time -> Edge Denoising -> Latent SMOTE -> Evidence Fusion -> Bayes Head -> Conformal CRC*

## TIER 1: SPATIO-TEMPORAL NEURAL ENCODING & ADVERSARIAL DENOISING

### Module 1: Continuous Temporal Encoder

**Tri-Band Harmonic Position Pos(t):**




**Self-Exciting Hawkes Rate $\lambda_u(t)$:**




**Output Tensor:** $\mathbf{h}_u^{(0)} \in \mathbb{R}^d$ **(Temporal Seed)**

### Module 2: Edge-Trust Filter & GNN

**Learnable Edge-Trust Gating Mechanism:**




**Spatio-Temporal Attention Aggregation:**

$$\tilde{\mathbf{h}}_u = \sum_{v \in \mathcal{N}_u} \alpha_{vu} \mathbf{W}_v \mathbf{h}_v^{(l-1)} \hat{g}_{vu}$$

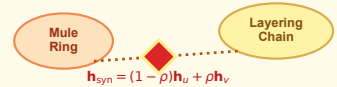
**Dirichlet Energy Preserved:** $\mathcal{E}(\mathbf{H}) > 0.75$

Threshold Floor: $\delta_{\text{floor}} = 0.10$

**Structural Embedding:** $\mathbf{h}_u^{(L)} \in \mathbb{R}^d$

### Module 3: Latent GraphSMOTE

**Latent Typology Manifold Synthesis:**




**Bilinear Link Gen & Physical Bounds:**

$$\hat{A}_{\text{syn},w} = \sigma(\mathbf{h}_{\text{syn}}^T \mathbf{W}_e \mathbf{h}_w)$$

**Physical Invariant Guardrail:** $\Phi_{\text{flow}} \approx 1.0$

*Offline Training Quarantined (0% Test Leak)*

**Balanced Training Graph:** $\tilde{\mathcal{G}}_{\text{train}}$ **(+9.92 pp Lift)**

**Offline Augmentation**

**Structural Embedding** $\mathbf{h}_u^{(L)} \in \mathbb{R}^d$ | **12-D Canonical Invariants** $\mathbf{z}_{\text{inv}} \in \mathbb{R}^{12}$ **(Hydrodynamic Mass-Flow + Hawkes + Taint)** | **Edge Gates** $\hat{g}_{vu} \geq 0.10$

## TIER 2: EVIDENCE FUSION, BAYESIAN OPTIMIZATION & CONFORMAL RISK CONTROL

### Module 4: Evidence-Adaptive Fusion

**Adaptive Inductive Gating Unit** $\alpha_u \in [0, 1]$**:**

$$\mathbf{h}_u^* = \alpha_u \mathbf{h}_u^{(L)} + (1 - \alpha_u)\mathbf{z}_{\text{inv}}$$

where $\alpha_u = \sigma(\mathbf{w}_\alpha^T[\mathbf{h}_u^{(L)} \| \mathbf{z}_{\text{inv}}])$

**Cold-Start (deg <= 1)**
$\alpha_u \to 0$ (Inv Dominates)
**89.60% Cold-Start F1**

**Dense Graph (deg >> 1)**
$\alpha_u \to 1$ (GNN Dominates)
**Multi-Hop Topology**

**12-D Canonical Invariant Inputs:**

$\Phi_{\text{flow}}$ **(Kirchhoff Mass-Flow) +** $\lambda_u$ **(Hawkes)**

$\mathbf{s}_{\text{fwd}}, \mathbf{s}_{\text{bwd}}$ (Causal Taints) + Moments $(\mu, \sigma^2, \gamma, \kappa)$

**Zero Retraining across Heterogeneous Ledgers**

**Fused Evidence Vector:** $\mathbf{h}_u^* \in \mathbb{R}^d$

### Module 5: Cost-Sensitive Bayes Head

**Bayesian Cost-Sensitive Risk Optimization:**

$$\mathcal{L}_{\text{asym}} = 15 \cdot \mathbb{I}(y=1)\log\hat{p} + 1 \cdot \mathbb{I}(y=0)\log(1-\hat{p})$$

15:1 Asymmetry: $150k FinCEN Penalty vs $10k Review Cost




**Multi-Task Representation Loss:**

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{asym}} + \lambda_1 \mathcal{L}_{\text{edge}} + \lambda_2 \mathcal{L}_{\text{flow}}$$

**Non-Conformity Scoring:** $s_i = 1 - \hat{p}_i(y)$

*Calibrated Posterior for Distribution-Free CRC*

**Calibrated Risk Score:** $\hat{p}_u \in [0, 1]$

### Module 6: Conformal CRC Arbiter

**Class-Conditional Quantiles** $(1 - \alpha \geq 99\%)$**:**

$$\hat{q}^{(0)} = \text{Quantile}\left(s_i^{(0)}, \frac{\lceil (n_0+1)(1-\alpha_0) \rceil}{n_0}\right)$$

$$\hat{q}^{(1)} = \text{Quantile}\left(s_i^{(1)}, \frac{\lceil (n_1+1)(1-\alpha_1) \rceil}{n_1}\right)$$

**Tier 1: Quarantine Hold** **0.18% (18/10k)**
$\Gamma(u) = \{1\}$ | Auto Wire Freeze & FinCEN Form 111

**Tier 2: Compliance Desk** **0.49% (49/10k)**
$\Gamma(u) = \{0, 1\}$ | **-99.51% Human Review Queue Cut**

**Tier 3: Straight-Through (STP)** **99.33% (9.9k/10k)**
$\Gamma(u) = \{0\}$ | **0.45 ms Real-Time Clear (<10ms SLA)**

**Deterministic Decision Set:** $\Gamma(u) \subseteq \{0, 1\}$